

PERC Admission Operations Engine

System Architecture Proposal (For Technical Review)

Purpose

This document defines the proposed architecture for the PERC Admission Operations Engine. The objective is to align the development team on how the system should be designed before implementation begins.

This document focuses on:

- Overall architecture
- Engine responsibilities
- Inputs and outputs
- Communication between engines
- Event flow
- Orchestration strategy
- Technology recommendations

Vision

The product is **not a CRM** and **not an AI chatbot**.

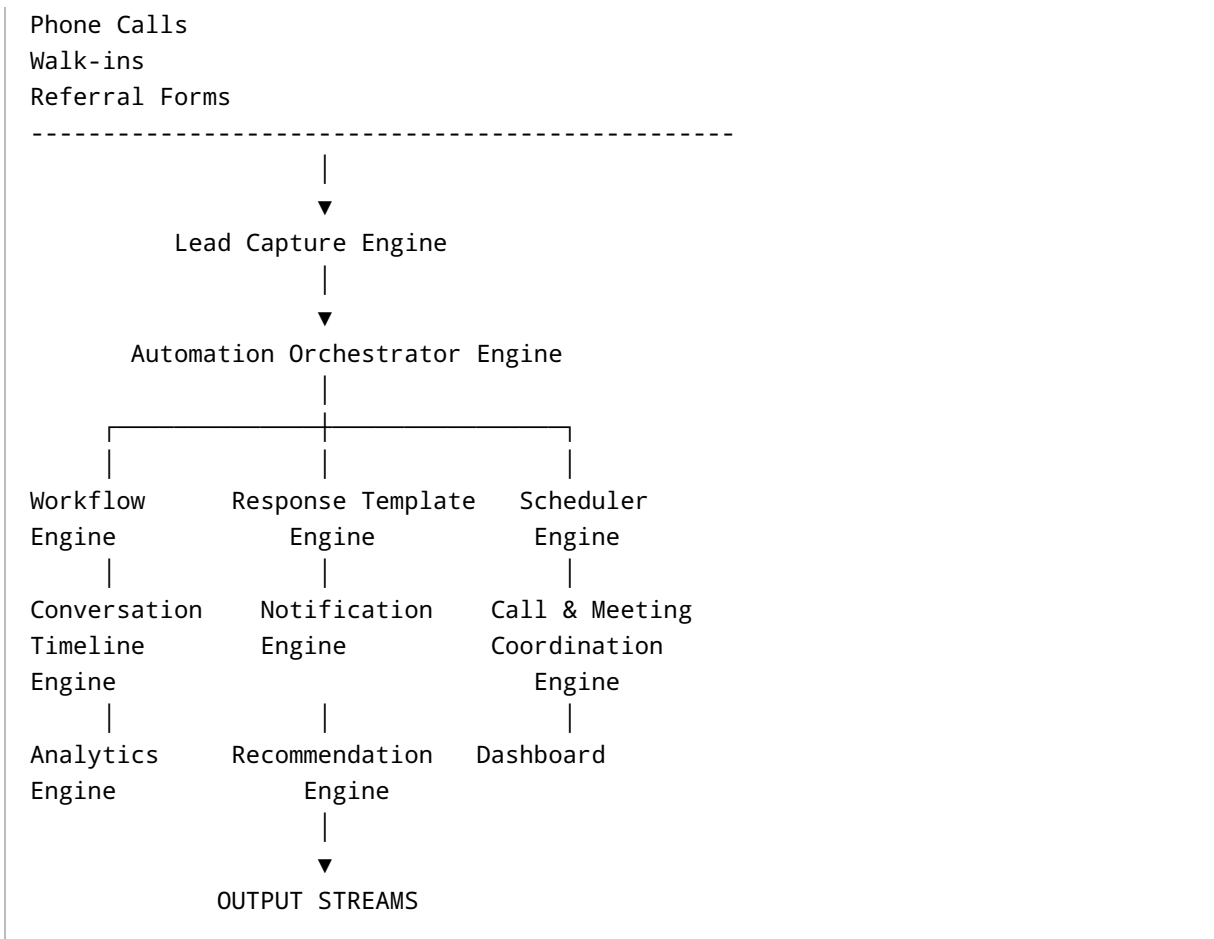
It is an **Admission Operations Engine** whose primary objective is:

Capture every enquiry, automate repetitive operational work, assist administrators, and ensure that no potential admission is lost due to manual delays or missed follow-ups.

High-Level Architecture

INPUT STREAMS

Website
WhatsApp
Instagram
Facebook
Google Business
Email



Core Architectural Principle

Every enquiry becomes a **Workflow Instance**.

Each workflow owns:

- Lead Information
- Current State
- Conversation Timeline
- Scheduled Tasks
- Meetings
- Documents Shared
- Analytics
- Recommendations

The workflow continues until:

- Admission Completed

- Lead Closed
 - Lead Lost
-

Engine Responsibilities

Engine 1 – Lead Capture Engine

Purpose

Capture enquiries from every supported source and normalize them into a standard lead format.

Inputs

- Website Forms
- WhatsApp
- Instagram
- Facebook
- Google Business
- Email
- Phone Calls
- Walk-ins

Responsibilities

- Capture enquiry
- Validate data
- Generate Lead ID
- Identify enquiry source
- Store timestamp
- Store contact information
- Detect enquiry category
- Create lead record
- Initialize workflow
- Trigger "Lead Created" event

Outputs

- Lead Created
 - Workflow Initialized
 - Lead Metadata
-

Engine 2 – Response Template Engine

Purpose

Select and prepare standardized responses.

Inputs

- Lead Information
- Enquiry Category
- Conversation Context

Responsibilities

- Select message template
- Attach brochure
- Attach course details
- Attach fee details
- Attach location details
- Prepare response
- Deliver through configured communication channel

Outputs

- Response Delivered
 - Delivery Status
-

Engine 3 – Workflow Engine

Purpose

Maintain the lifecycle of every lead.

Inputs

- Lead Events
- Admin Actions
- Lead Responses

Responsibilities

- Update workflow state
- Validate state transitions
- Pause workflow
- Resume workflow
- Close workflow

- Recover inactive workflow

Outputs

- Workflow Updated
 - State Changed Event
-

Engine 4 – Scheduler Engine

Purpose

Execute delayed and scheduled operations.

Inputs

- Scheduling Requests
- Meeting Requests
- Reminder Requests

Responsibilities

- Schedule follow-ups
- Schedule reminders
- Cancel schedules
- Reschedule jobs
- Retry failed jobs

Outputs

- Scheduled Event Triggered
 - Schedule Completed
-

Engine 5 – Conversation Timeline Engine

Purpose

Maintain complete interaction history.

Inputs

- Messages
- Calls
- Meetings
- Workflow Updates
- Admin Actions

Responsibilities

- Log every interaction
- Maintain chronological timeline
- Provide conversation context

Outputs

- Updated Timeline
-

Engine 6 – Follow-up Engine

Purpose

Ensure no enquiry becomes inactive without action.

Inputs

- Scheduler Events
- Workflow State
- Conversation Timeline

Responsibilities

- Evaluate inactivity
- Determine follow-up necessity
- Send follow-up
- Cancel unnecessary follow-ups
- Trigger recovery process

Outputs

- Follow-up Sent
 - Recovery Initiated
-

Engine 7 – Call & Meeting Coordination Engine

Purpose

Manage appointments between leads and administrators.

Inputs

- Meeting Requests
- Admin Availability

- Lead Availability

Responsibilities

- Find available slots
- Schedule meetings
- Reschedule missed meetings
- Send reminders
- Collect meeting feedback

Outputs

- Meeting Scheduled
 - Meeting Updated
 - Meeting Completed
-

Engine 8 – Notification Engine

Purpose

Notify administrators of important operational events.

Inputs

- System Events
- Workflow Events
- Scheduler Events

Responsibilities

- Dashboard notifications
- Daily summaries
- Critical alerts
- Escalation notifications

Outputs

- Notifications Delivered
-

Engine 9 – Analytics Engine

Purpose

Generate operational insights.

Inputs

- Lead Events
- Workflow Events
- Meeting Events
- Admissions

Responsibilities

- Update metrics
- Calculate conversions
- Generate reports
- Maintain dashboards

Outputs

- Updated Analytics
 - Reports
-

Engine 10 – Recommendation Engine

Purpose

Suggest the next best action to administrators.

Inputs

- Workflow State
- Timeline
- Meetings
- Follow-up Status

Responsibilities

- Prioritize leads
- Recommend calls
- Recommend meetings
- Recommend recovery actions

Outputs

- Next Best Action
-

Engine 11 – Automation Orchestrator Engine

Purpose

Coordinate every engine and control workflow execution.

The Orchestrator **does not perform business logic**. It coordinates the execution of specialized engines.

Inputs

- New Lead Events
- Scheduler Events
- Lead Replies
- Admin Actions
- Meeting Events
- Payment Events
- Workflow Events

Responsibilities

- Receive events
- Evaluate workflow conditions
- Decide next actions
- Invoke required engines
- Execute independent tasks in parallel
- Wait only for dependent tasks
- Handle failures
- Retry operations
- Cancel obsolete tasks
- Maintain workflow consistency

Outputs

- Engine Execution Requests
- Workflow Decisions
- New System Events

Communication Model

The architecture should follow an **Event-Driven** approach.

The Orchestrator receives an event, evaluates it, and coordinates the required engines.

Example:

New Lead



Lead Capture Engine



Automation Orchestrator



Timeline Engine

Analytics Engine

Notification Engine

(Executed in parallel)



Response Template Engine



Scheduler Engine



Workflow Updated

Parallel vs Sequential Execution

Sequential (Avoid)

Lead Created



Timeline



Analytics

↓

Notification

↓

Response

↓

Scheduler

Each engine waits for the previous one.

Higher latency.

Parallel (Recommended)

Lead Created

↓

Automation Orchestrator

↓

Timeline

Analytics

Notification

(All execute simultaneously)

↓

Response Engine

↓

Scheduler

Only dependent tasks wait.

Technology Recommendation

Frontend

- Next.js
- Tailwind CSS
- shadcn/ui

Purpose:

Internal Admin Dashboard

Backend

- NestJS
- TypeScript

Purpose:

Automation Orchestrator

Business Logic

REST APIs

Scheduler

Workflow Management

Database

Supabase PostgreSQL

Stores:

- Leads
- Workflows
- Templates
- Timeline
- Meetings
- Analytics

Storage

Supabase Storage

Stores:

- Brochures
- PDFs
- Documents
- Images

Authentication

Supabase Auth

Roles:

- Admin
- Counselor
- Teacher
- Student (Future)

Communication

WhatsApp Business Platform

Primary communication channel

Deployment

Frontend

Vercel

Backend

Google Cloud Run

Database

Future Scalability

The architecture should be modular so that new engines can be added without changing the overall system.

Examples:

- AI Recommendation Engine
 - Payment Engine
 - Student Enrollment Engine
 - Attendance Engine
 - Marketing Automation Engine
 - CRM Integration Engine
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Key Design Principles

1. Every enquiry becomes a workflow.
2. Every engine has a single responsibility.
3. The Orchestrator coordinates but does not contain business logic.
4. Independent tasks execute in parallel whenever possible.
5. Sequential execution is used only when a business dependency exists.
6. Every interaction is recorded in the timeline.
7. Every workflow is event-driven.
8. Administrators intervene only where human judgment is required.
9. Automation should reduce repetitive manual work while maintaining full operational visibility.
10. The architecture must remain modular, scalable, and maintainable.